

Lesson 62 Practice: Current Internet, Networking and Cloud IaaS

JKSSB Computer Awareness

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Provenance: All 20 questions are original JKSSB-pattern questions (*[Original]*); concepts drawn from PYQ-18/PYQ-19/PYQ-20; none reproduces an official paper item.

Q1. *[Original]* Expand TCP.

- A. Transfer Control Program
- B. Transmission Control Protocol
- C. Transport Communication Process
- D. Tele Communication Protocol

Q2. *[Original]* Which statement correctly describes TCP delivery guarantees?

- A. TCP guarantees ordered and reliable delivery, but provides no timing guarantee.
- B. TCP guarantees delivery within a fixed time but not the order.
- C. TCP delivers datagrams without setup and without retransmission.
- D. TCP transfers only control messages and uses no acknowledgements.

Q3. *[Original]* Which pairing of transport protocol and property is correctly matched?

- A. UDP — guarantees ordered delivery of datagrams
- B. TCP — connectionless with no retransmission
- C. UDP — connectionless, fast, with no ordering or reliability guarantee
- D. TCP — transfers only diagnostic messages using no ports

Q4. *[Original]* Expand ICMP.

- A. Internet Control Message Protocol
- B. Internal Computer Mail Protocol
- C. Internet Communication Management Process
- D. Inter-Connected Media Protocol

Q5. *[Original]* Which of the following is **NOT** true of ICMP?

- A. It carries control and diagnostic messages about the network.
- B. It supports tools such as ping and traceroute.
- C. It transfers application data such as files and web pages.
- D. It reports errors such as destination unreachable.

Q6. *[Original]* Expand BGP.

- A. Basic Gateway Process
- B. Border Gateway Protocol
- C. Broadband Group Protocol
- D. Binary Gateway Program

Q7. *[Original]* BGP is best described as:

- A. an inter-domain routing protocol that routes between autonomous systems.
- B. a protocol for routing inside a single small local network.
- C. a mail-transfer protocol that downloads mail to the local device.
- D. a diagnostic protocol that carries no routing information.

Q8. *[Original]* Which statement correctly distinguishes BGP from OSPF and RIP?

- A. BGP routes between autonomous systems; OSPF and RIP route inside a single domain.
- B. OSPF routes between autonomous systems; BGP routes inside one LAN.
- C. RIP is inter-domain; BGP and OSPF are mail protocols.
- D. All three transfer application data over ports.

Q9. *[Original]* IPv6 addresses are:

- A. 32-bit long.
- B. 64-bit long.
- C. 128-bit long.
- D. 256-bit long.

Q10. *[Original]* Which statement about IPv6 is correct?

- A. IPv6 uses a fixed 40-byte header and has no broadcast address.
- B. IPv6 uses a fixed 20-byte header and relies on broadcast delivery.
- C. IPv6 uses 256-bit addresses with a broadcast address for all nodes.
- D. IPv6 is a routing protocol that runs between autonomous systems.

Q11. *[Original]* The reason IPv6 removes the need for NAT is:

- A. its vastly larger 128-bit address space.
- B. its use of TCP instead of UDP.
- C. its support for broadcast delivery.
- D. its ability to transfer application data.

Q12. *[Original]* HTTP/3 differs from HTTP/1.1 and HTTP/2 in that it:

- A. runs over TCP like the earlier versions.
- B. runs over QUIC over UDP instead of TCP.
- C. transfers only control messages with no ports.

D. routes between autonomous systems on the backbone.

Q13. *[Original]* Expand DNS.

- A. Domain Name System
- B. Data Network Service
- C. Digital Number Sequence
- D. Direct Node Switch

Q14. *[Original]* Which statement correctly describes DNS?

- A. It translates domain names into IP addresses through a hierarchy of servers.
- B. It routes packets between autonomous systems on the backbone.
- C. It downloads mail from the server to the local device.
- D. It encrypts message bodies for confidentiality.

Q15. *[Original]* Match each Internet-governance actor to its role:

- A. PIR — operates the .org registry; ICANN — coordinates domain names and IP addresses globally.
- B. PIR — coordinates all IP addresses; ICANN — operates the .org registry.
- C. PIR — routes between autonomous systems; ICANN — transfers mail.
- D. PIR — assigns TCP ports; ICANN — sends diagnostic messages.

Q16. *[Original]* ARPANET is best described as:

- A. the 1969 packet-switched ancestor of the Internet.
- B. a circuit-switched telephone network with dedicated calls.
- C. a cloud service model managing networking hardware.
- D. a mail protocol that syncs mail across devices.

Q17. *[Original]* Which statement correctly distinguishes packet switching from circuit switching?

- A. Packet switching splits data into addressed packets taking possibly different routes; circuit switching holds one dedicated path.
- B. Circuit switching splits data into packets; packet switching holds a dedicated circuit.
- C. Both hold a dedicated path for the whole session.
- D. Both deliver packets with guaranteed timing.

Q18. *[Original]* Consider the assertion and reason about a router. Assertion (A): A router connects different networks and forwards packets between them. Reason (R): A router reads the destination IP address and chooses the next hop from its routing table.

- A. Both A and R are true, and R is the correct explanation of A.
- B. Both A and R are true, but R is not the correct explanation of A.
- C. A is true but R is false.
- D. A is false but R is true.

Q19. *[Original]* Evaluate the statements:

- I. TCP guarantees ordered and reliable delivery but no timing guarantee.
 - II. ICMP transfers application data such as files and web pages.
 - III. In IaaS, the provider manages the networking hardware while the customer manages the operating system, applications, and data.
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- A. I and III only
 - B. II only
 - C. I, II, and III
 - D. I and II only

Q20. *[Original]* Which pairing is correctly matched?

- A. BGP — routes inside one LAN; ICMP — transfers files
- B. HTTP/3 — uses QUIC over UDP; IPv6 — 128-bit with no broadcast
- C. IaaS provider — manages the customer OS and data; PIR — coordinates TCP ports
- D. UDP — guarantees order; TCP — connectionless and fast